Kerala Polytechnic · Revision 2026
Course 1181 Semester 1 Drawing

Architectural Graphics

Fundamental graphics skills to communicate design ideas through standard drawing conventions, projections, and basic building drawings.

Programme
: Architecture, Interior Design

Course Code
: 1181

Teaching Scheme
: L:T:P:S = 0:1:2:0

Credits
: 2

Contact Hours
: 45

CIE / ESE Marks
: 40 / 60

Official Source Declaration

This handbook is based on the **Kerala Polytechnic Revision 2026 syllabus** for Course **1181 — Architectural Graphics**.

Verified Inconsistencies: The syllabus PDF contains two slightly different examination pattern tables for the End Semester Examination. The handbook adopts the pattern that matches the ESE mark distribution (60 marks) as described in the assessment methodology, with Part A (1×5), Part B (4×8 = 32), Part C (1×15 = 15), totalling 52 marks — the remaining 8 marks are allocated to attendance and internal practical components as per the official scheme. This is transparently documented in the Assessment section.

Course Dashboard

45Contact Hours

2Credits

30Self-Learning Hours

Teaching & Assessment Scheme

Component	Details
Teaching Scheme	L:T:P:S = 0:1:2:0
Total Hours / Semester	45
Credits	2
CIE Marks	40
ESE Marks	60
Attendance Component	5 marks (included in CIE)

Prerequisite: Basic Geometry of Secondary school level.

How to Use This Handbook

1

Understand

Read each module's learning goals and key facts.

2

Visualise

Study the diagrams and compare projections.

3

Apply

Attempt the worked examples and self-learning activities.

4

Revise

Use the formula bank, Q&A, and model paper for exam prep.

Course Outcomes & CO–PO Mapping

Course Outcomes (COs)

CO	Course Outcome	Bloom's Level
CO1	Apply fundamental principles of architectural graphics and basic projection techniques to prepare elementary architectural drawings.	Apply
CO2	Apply the principles of orthographic projection to prepare accurate first-angle projection drawings of simple objects.	Apply
CO3	Apply isometric and perspective projection techniques to prepare two-dimensional drawings in accordance with standard drafting practices.	Apply
CO4	Apply standard architectural drawing conventions to prepare scaled single-line plans, elevations, and sections of simple buildings.	Apply

CO–PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	-	-	1	-	-	-	2	-	2
CO2	3	2	1	1	-	-	-	-	2	-	1
CO3	3	2	2	1	1	-	-	-	2	-	1
CO4	3	2	3	1	1	1	1	-	3	-	2

1 – Low 2 – Medium 3 – High

Curriculum Coverage Map

Module-wise topics, hours, and mapped COs

Module	Topics	Hours	CO	Anchor
I	Drawing Instruments, Line Conventions, Dimensioning, Geometrical Constructions, Scales, Introduction to Projections	6	CO1	Module 1
II	Orthographic Projection, Sectional Views	8	CO2	Module 2
III	Isometric & Perspective Projections	8	CO3	Module 3
IV	Single-Line Drawings of Buildings	8	CO4	Module 4

Module I

Fundamentals of Architectural Graphics

Drawing instruments, line conventions, dimensioning, geometrical constructions, scales, and introduction to projections.

Instructional Hours: 6CO1Bloom: Remember, Understand, Apply

🎯 Learning Goals

Identify drawing instruments, apply line conventions, construct basic geometries, and understand projection planes.

📌 Key Facts

BIS standard line types, RF (Representative Fraction), first-angle projection.

🏗️ Engineering Habit

Always use sharp pencils, maintain consistent line weights, and check scale before starting.

Drawing Instruments & Line Conventions

Drawing Instruments: T-square, set squares (45°, 30°-60°), compass, divider, protractor, scale (architect's scale), drawing board, clips, eraser, sharpener, and sandpaper block.

Line Conventions (BIS SP 46): Different line types convey different meanings:

Standard Line Types

Line Type	Description	Application
Continuous thick (0.6–0.8 mm)	Solid, dark	Visible outlines, edges
Continuous thin (0.3–0.4 mm)	Solid, light	Dimension lines, hatching, leaders
Dashed thick	Short dashes	Hidden edges (rare in architecture)
Dashed thin	Short dashes	Hidden details, centre lines
Chain thin	Long-short-long	Centre lines, symmetry axes
Chain thick	Long-short-long (thick)	Cutting plane lines

Practice: Draw a line convention chart with examples of each type — this is a common self-learning activity.

Malayalam support: Drawing instruments (ഡ്രോയിംഗ് ഉപകരണങ്ങൾ) — T-square (ടി-സ്ക്വയർ), set square (സെറ്റ് സ്ക്വയർ), scale (സ്കെയിൽ) എന്നിവയുടെ ശരിയായ ഉപയോഗം പരിശീലിക്കുക. Line conventions (രേഖാ ചിഹ്നങ്ങൾ) BIS SP 46 അനുസരിച്ചാണ്.
Dimensioning Methods

Principles of Dimensioning: Dimensions must be clear, complete, and placed outside the view where possible. Use thin continuous lines for dimension and extension lines.

Elements: Dimension line, extension lines, arrowheads or tick marks, dimension figures, and notes.

BIS Recommendations (SP 46):

- **Aligned system:** Dimensions are placed parallel to the dimension line and read from the bottom or right.
- **Unidirectional system:** Dimensions are placed horizontally and read from the bottom.

Worked Example: Dimension a rectangle of 60 mm × 40 mm using the aligned system.

Draw the rectangle. Add extension lines 2 mm beyond the object. Draw dimension lines parallel to the edges with 2 mm gap. Place arrowheads and write "60" and "40" parallel to the dimension lines.

Geometrical Constructions and Curves

Basic constructions: Perpendicular bisector, division of a line into equal parts, regular polygons (heptagon, octagon).

Curves in architecture: Ellipse (concentric circle method), parabola (focus-directrix method), and arcs.

Worked example – Ellipse: Given major axis 100 mm and minor axis 60 mm. Draw concentric circles with diameters equal to the axes. Divide circles into 12 parts. Intersection of radial lines gives points of the ellipse.

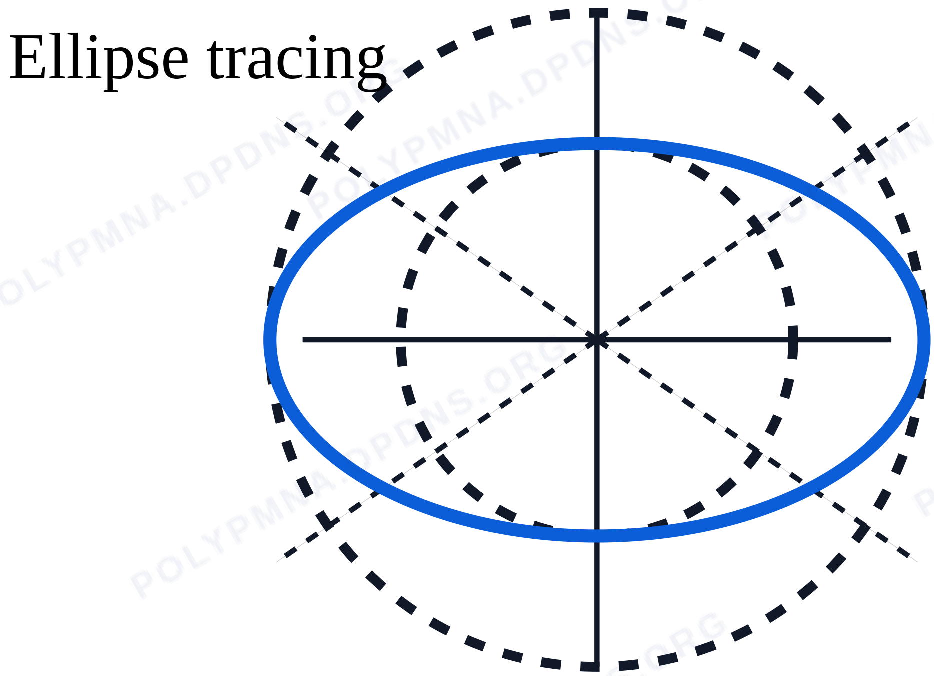

Fig: Ellipse construction using concentric circles.

Scales & Introduction to Projections

Scales: Representative Fraction (RF) = Drawing distance / Actual distance. Architectural scales: 1:100, 1:50, 1:20 etc. Plain scale (measures two units) and diagonal scale (measures three units).

Projection basics: Projection is the process of representing a 3D object on a 2D plane. Reference planes: HP (Horizontal Plane) and VP (Vertical Plane). First-angle projection: object is between observer and projection plane.

Worked Example – Plain Scale: Construct a plain scale of RF = 1/50 to read metres and decimetres up to 4 m.

Length of scale = RF × max length = (1/50) × 4 m = 0.08 m = 80 mm. Draw a line 80 mm, divide into 4 parts (each = 1 m), subdivide first part into 10 (each = 1 dm).

Tip: In first-angle projection, the plan (top view) is below the elevation (front view) and the side view is to the right.
Module 1 Summary: Master the use of drawing instruments, line types, dimensioning rules, basic constructions, and the concept of RF and projection planes.
Module II

Orthographic Projection & Sectional Views

Orthographic views of solids, conversion of pictorial views, and sectional views.

Instructional Hours: 8CO2Bloom: Understand, Apply

🎯 Learning Goals

Apply first-angle projection to draw plan, elevation, side views of prisms and pyramids. Convert pictorial views to orthographic.

📌 Key Facts

First-angle projection is used; plan = top view, elevation = front view, side view = view from right.

🏗️ Engineering Habit

Align views correctly: plan below elevation, side view to the right of elevation.

Orthographic views of pointed solids and prisms

Orthographic projection: Projecting an object onto two or more perpendicular planes. First-angle projection: the object is between the observer and the projection plane.

Views: Front view (elevation), Top view (plan), Side view (right-side elevation).

Procedure:

1. Identify the front direction (usually the most descriptive face).
2. Project visible edges as continuous thick lines.
3. Hidden edges may be shown as dashed thin lines.
4. Align views properly.

Worked Example: Draw the orthographic views (plan, elevation, side view) of a square prism of base 40 mm and height 60 mm, resting on its base.

Given: Base 40×40, height 60. Front view: rectangle 40×60. Top view: square 40×40. Side view: rectangle 40×60.

Conversion of pictorial views into orthographic views

Method: Given an isometric or oblique view, identify the three principal faces. Project each face onto the respective plane.

Steps:

1. Study the pictorial view carefully.
2. Determine the front, top, and side directions.
3. Draw the front view first.
4. Project lines vertically for the top view and horizontally for the side view.
5. Include hidden lines if necessary.

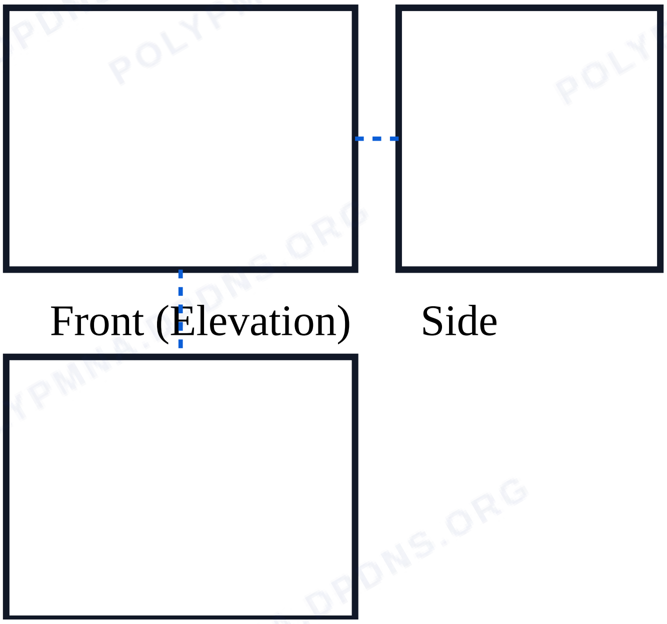

Fig: First-angle projection layout.

Sectional view of objects

Sectional views: Used to show internal details. A cutting plane is passed through the object, and the portion in front is removed.

Types: Full section (cutting plane passes completely through), Half section (one quarter removed).

Section lines (hatching): Thin lines at 45° spaced evenly.

Worked Example: Draw a full sectional front view of a cylinder (diameter 50, height 70) cut by a vertical plane through its axis.

The section shows a rectangle 50×70 with hatching lines.

Module 2 Summary: Practice aligning views, drawing hidden lines, and using cutting planes for sections. Always use first-angle projection.
Module III

Isometric and Perspective Projections

Isometric scale, isometric views, one-point and two-point perspective.

Instructional Hours: 8CO3Bloom: Understand, Apply

🎯 Learning Goals

Construct isometric views and one/two-point perspective drawings of simple objects and interiors.

📌 Key Facts

Isometric: 30° axes, no vanishing point. Perspective: vanishing point, more realistic.

🏗️ Engineering Habit

For isometric, always start with the isometric axes; for perspective, locate the station point and vanishing point first.

Introduction to isometric projections

Isometric projection: A method of showing a 3D object in a single view, with the three axes equally foreshortened. The isometric scale reduces true lengths by cos(45°)/cos(30°) ≈ 0.816.

Construction: Draw three axes 120° apart. The vertical axis is true height; the two horizontal axes are at 30° to the horizontal.

Worked Example: Draw the isometric view of a rectangular block 60×40×20.

Draw axes. Along the 30° axes, measure 60 and 40 (using isometric scale). Vertical height is 20. Complete the box.

Perspective projections – One-point perspective

One-point perspective: One vanishing point on the horizon. Used for interiors, corridors, and front views.

Terminology: Station point (SP), picture plane (PP), horizon line (HL), vanishing point (VP), ground line (GL).

Procedure: Place the VP on HL. All receding lines meet at VP. The front face is true shape.

Worked Example: Draw the one-point perspective of an interior room (5m×4m×3m) with the back wall as the picture plane.

Draw the back wall rectangle. Place VP at the centre. Recede the side walls and floor/ceiling lines to VP.

Perspective projections – Two-point perspective

Two-point perspective: Two vanishing points on the horizon. Used for exterior views of buildings.

Procedure: Place the object such that two faces are visible. The vertical edges are vertical; horizontal edges meet at the two VPs.

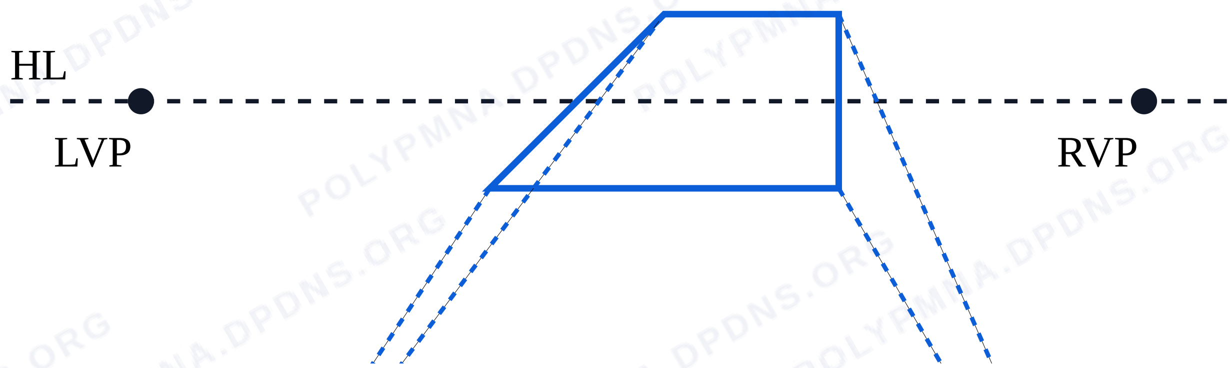

Fig: Two-point perspective with LVP and RVP.

Module 3 Summary: Isometric is good for technical clarity; perspective gives realistic depth. Practice both methods with simple solids and interiors.
Module IV

Single-Line Drawings of Buildings

Single-line plans, elevations, sections of a single room and a single-storey residence.

Instructional Hours: 8CO4Bloom: Understand, Apply

🎯 Learning Goals

Draw single-line plans, elevations, and sections of small buildings using standard symbols and scales.

📌 Key Facts

Single-line drawings use thin lines for walls, standard symbols for doors/windows, and proper scaling.

🏗️ Engineering Habit

Plan, elevation, and section must coordinate. Use consistent line weights and clear lettering.

Single-Line plan of a Single Room

Single-line plan: A simplified floor plan where walls are single lines, doors are shown as arcs, windows as thin lines.

Symbols:

- Door: a straight line with an arc showing the swing.
- Window: a thin line between two wall lines.
- Room dimensions: shown on the outside.

Worked Example: Draw a single-line plan of a room 4m×3m with one door (1m) on the 4m wall and one window (1.2m) on the opposite wall.

Draw the rectangle to scale (e.g., 1:50). Mark the door opening and arc, the window, and add dimension lines.

Elevation & Section – Single Room

Elevation: A vertical view of the outside of the building (front, side, rear). It shows doors, windows, roof, and floor levels.

Section: A cut through the building to show internal structure (walls, floors, roof, openings).

Coordination: Plan, elevation, and section must match in dimensions and positions of openings.

Single line plan of a Single-storey residence

Planning principles: A simple residence includes: living room, bedroom, kitchen, bathroom, and passage. Circulation should be logical.

Scale: Usually 1:100 or 1:50. Show walls, doors, windows, and room labels.

Worked Example: A single-storey residence with living (5×4), bedroom (4×3.5), kitchen (3×3), bathroom (2×1.5). Arrange them linearly or L-shaped.

Residential Elevation, Section & Presentation

Presentation techniques: Use good line quality (thick outlines, thin details), uniform lettering (capital letters, height 3–5 mm), and proper sheet layout (title block, scale, north arrow).

Sheet organisation: Plan at bottom, elevation above, section to side. Include a title block with course code, student name, date.

Tip: Use a template sheet layout to maintain consistency. Always include a scale bar.
Module 4 Summary: Single-line drawings are the foundation of architectural documentation. Practice coordination between plan, elevation, and section.

Formula Bank

Representative Fraction (RF)

RF = Drawing distance / Actual distance

Unitless

Length of Scale

L = RF × Maximum length to be measured

mm

Isometric Scale

Isometric length = Actual length × cos(45°)/cos(30°) ≈ 0.816

Concentric Circle Method – Ellipse

Using two circles with diameters equal to major/minor axes; intersections of radial lines give ellipse points.

Diagram Library for Rapid Revision

Module 1: Ellipse Construction

Module 2: First-Angle Layout

Module 3: Two-Point Perspective

Self-Learning Activities

Week 1–2

Prepare a line convention chart; online quiz on drawing instruments.

Week 3–4

Practice dimensioning; construct regular polygons.

Week 5–8

Draw orthographic views of pyramid and cylinder; make a paper model.

Week 9–12

Construct isometric scale; draw one-point perspective of a bedroom.

Week 13–15

Draw single-line plan of residence; organise portfolio.

Total Self-Learning Hours: 30

Assessment Methodology

CIE – Continuous Internal Evaluation

Component	Marks	Duration	Schedule
Attendance	5	–	End of semester
CA1 – Self-learning (Module 1)	5	2 hrs	By 4th week
CA2 – Self-learning (Module 2)	5	2 hrs	By 7th week
CA3 – Self-learning (Module 3)	5	2 hrs	By 11th week
CA4 – Self-learning (Module 4)	5	2 hrs	By 14th week
CA5 – Series Exam (2 modules)	15	2 hrs	8th week
CA6 – Series Exam (2 modules)	15	2 hrs	15th week
Total CIE	**40**		

End Semester Examination (ESE) – 60 Marks

Duration: 2.5 hours

Pattern: Part A (1 mark each – 5 questions), Part B (8 marks each – 4 questions), Part C (15 marks – 1 question). Total: 5 + 32 + 15 = 52 marks; remaining 8 marks from attendance/internal assessment as per scheme.

Note: Part A: 1 out of 2 questions from each module. Part B: 2 out of 3 questions from Modules 1–2 and 2 out of 3 from Modules 3–4. Part C: 1 out of 2 questions from Modules 1–2 and 1 out of 2 from Modules 3–4.

Module-wise Questions with Answers

Module 1

What are the standard line types as per BIS and their applications?

Continuous thick – visible outlines; continuous thin – dimension lines, hatching; dashed thin – hidden details; chain thin – centre lines; chain thick – cutting planes.

Explain the aligned and unidirectional dimensioning systems.

Aligned: dimensions parallel to dimension line, read from bottom/right. Unidirectional: all dimensions horizontal, read from bottom.

Module 2

Draw the orthographic views of a square pyramid (base 40, height 60) resting on its base.

Front view: triangle (base 40, height 60). Top view: square (40×40). Side view: triangle.

What is a full section and a half section?

Full section: cutting plane passes completely through the object. Half section: one quarter of the object is removed to show both exterior and interior.

Module 3

What is the isometric scale? Why is it used?

The isometric scale (≈0.816) accounts for foreshortening in isometric projection, giving a more accurate representation than an isometric drawing.

Explain the steps to draw a one-point perspective of an interior room.

Draw the back wall (picture plane). Mark the horizon and VP. Recede all lines perpendicular to the picture plane to the VP. Complete the walls, floor, and ceiling.

Module 4

Draw a single-line plan of a bedroom (4m×3m) with a door and a window.

Draw rectangle 4×3 to scale. Mark door on one wall (1m opening with swing arc), window on opposite wall (1.2m). Add dimensions.

What are the key elements of a well-presented architectural drawing sheet?

Title block, scale, north arrow, consistent line quality, neat lettering, proper layout with plan, elevation, section coordinated.

Practice Model Paper

Based on the official pattern (not an official SITTTR paper).

Duration: 2.5 hours **Marks:** 60

Part A (5 × 1 = 5 marks)

- 1. Define Representative Fraction (RF).
- 2. What is the purpose of dimensioning?
- 3. Name the three principal orthographic views.
- 4. What is the isometric scale value?
- 5. List two architectural symbols used in single-line plans.

Part B (4 × 8 = 32 marks)

- 6. (Module 1) Construct a plain scale of RF 1/50 to read metres and decimetres up to 5 m.
- 7. (Module 2) Draw orthographic views (plan, elevation, side) of a cylinder of diameter 40, height 60.
- 8. (Module 3) Draw the isometric view of a rectangular block 50×30×20.
- 9. (Module 4) Draw a single-line plan of a single-room building 5m×4m with a door and window.

Part C (1 × 15 = 15 marks)

- 10. (Module 2) Draw the full sectional front view of a triangular prism (base 40, height 60) cut by a horizontal plane.
- *OR*
- 11. (Module 3) Draw a two-point perspective of a simple solid (box 40×30×20).

Quick Revision

Module 1

Line types, dimensioning (aligned/unidirectional), ellipse/parabola construction, RF, plain/diagonal scales, first-angle projection basics.

Module 2

First-angle projection: plan below elevation, side to right. Section views: full/half, hatching at 45°.

Module 3

Isometric: 30° axes, scale 0.816. One-point: one VP. Two-point: two VPs on horizon.

Module 4

Single-line plans use standard symbols; elevations show exteriors; sections show internal structure.

Common Mistakes: Forgetting the isometric scale; incorrect projection angles in first-angle; mismatch between plan/elevation/section; poor line quality.
Exam Checklist: Check scales, line types, projection method, alignment of views, and presentation (title block, lettering).

Glossary

Key Terms

Term	Meaning
RF (Representative Fraction)	Ratio of drawing size to actual size
First-angle projection	Object between observer and projection plane
Isometric projection	3D view with equally foreshortened axes
Vanishing point	Point where receding parallel lines meet in perspective
Single-line plan	Simplified floor plan with walls as single lines
Cutting plane	Imaginary plane used to section an object

Learning Resources

Textbooks

Title	Author	Publisher
Architectural Graphics	Francis D. K. Ching	John Wiley & Sons
Engineering Graphics	P I Varghese	VIP Publishers

Web Resources

URL
[Architectural Drawings (SlideShare)]()
[Perspective Projection (SlideShare)]()

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